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ARTICLE

Historical and Contemporary Gene Flow and the Genetic Structure of Muskellunge in the Ohio River Drainage

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Abstract
Muskellunge *Esox masquinongy* is a recreationally important species distributed throughout the Ohio, upper Mississippi, and Great Lakes drainages and has been introduced outside of its native range. We conducted a survey of microsatellite DNA variation among 11 populations, natural, managed, and hatchery, in the Ohio River drainage. Microsatellite DNA variation was highly significant among all populations. A CLUSTER analysis identified $K = 2$ populations corresponding to populations east and west of the Ohio River. This may be a consequence of different management practices by Ohio and West Virginia or may suggest that the Ohio River is a substantial barrier to gene flow. A comparison of historical and contemporary gene flow suggests little change in genetic exchange among populations. Our results further support the use of locally adapted broodstock in further management and supplementation of Muskellunge populations.

Historical (drainage changes, stream capture, glaciations) and contemporary (postnatal dispersal, ongoing gene flow, habitat loss) processes can contribute to a species' genetic population structure. For species with commercial or recreational value, management practices (population supplementation and interdrainage transfers) can significantly affect the species distribution and genetic population structure and can obscure historical patterns of structuring (e.g., White et al. 2012). The Muskellunge *Esox masquinongy* is an important game species with a native distribution in the upper Mississippi River, Ohio, Cumberland, and Tennessee rivers, Great Lakes, and the Hudson Bay drainage (Burr and Page 1986; Crossman and McAllister 1986; Starnes and Etnier 1986; Underhill 1986; Kerr 2011).

Three subspecies of Muskellunge were historically recognized based on color pattern and distribution (Crossman 1986): *E. masquinongy masquinongy* from the Great Lakes, *E. masquinongy ohioense* from the Ohio, Tennessee, and Cumberland rivers, and *E. masquinongy immaculatus* from the upper Mississippi River. The color patterns (clear, spotted, and barred) do not appear to show restricted or regional distributions (Hourston 1955). However, in a survey of allozyme variation Muskellunge samples grouped by drainage, which was consistent with subspecies distributions (Great Lakes, Ohio River, and upper Mississippi River; Koppelman and Philipp 1986).

Muskellunge are usually found at low densities (Miles 1978; Kapuscinski et al. 2013). There is evidence of spawning site fidelity in both lake and river systems (Crossman 1990; Younk et al. 1996; Wilson et al. 2016) despite considerable seasonal movements (LaPan et al. 1996). Certain populations also exhibit genetic and life history differences (Hanson et al. 1983, cited by Fronhauer et al. 2007; Younk and Strand 1992; Clapp and Wahl 1996; Wagner and Wahl 2007). It has been widely stocked both within and outside its native range, and interdrainage transfers have been common (Kapuscinski et al. 2013). In some states, hatchery programs are extensive (Clark 1964; Wingate and Younk 2007) and there are efforts to ensure the genetic integrity of source and target populations (Jennings et al. 2010).

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Our objective was to survey microsatellite DNA variation among populations of Muskellunge within the Ohio River drainage. We surveyed broodstock, managed, and natural populations in Ohio, Kentucky, and West Virginia. Since we know the histories of the broodstock populations, we examined possible contributions of these brood sources to current population structure. We estimated levels of historical and contemporary gene flow among these populations. Our results will provide recommendations for state and regional management of Muskellunge. This is the first study to examine genetic structure among Ohio River Muskellunge populations, a part of the distribution mostly lacking in a recent review of Muskellunge genetics (Miller et al. 2017).

METHODS

Population histories.— Some of the earliest stockings in Ohio were fry sourced from New York (Clark 1964). In the 1950s and 1960s stocking was widespread throughout Ohio using fish predominantly from the Ohio River strain, but some Great Lakes strain fish were used (Clark 1964). Current stocking practices in Ohio use fish from three sources, all of which are of Ohio River strain origin. Minor Clark State Fish Hatchery (locality 11; Table 1; Figure 1) uses fish from the Licking River (Kentucky), a tributary of the Ohio River. Fish from London State Fish Hatchery (locality 10) are taken from Clear Fork Reservoir, which was established with fish from several southern Ohio streams, including Sunfish Creek, Scioto Brush Creek, and Salt Creek. Kincaid State Fish Hatchery (locality 9) in Ohio obtains its broodstock from Salt Fork Lake, which was established from Clear Fork Reservoir fish (E. Heyob, Ohio Department of Natural Resources, personal communication). The three hatcheries provide advanced fingerlings that are used to stock 10 reservoirs in Ohio. Sunfish Creek (locality 3) has potentially made a substantial contribution to Ohio broodstock populations, and we would predict that it should show evidence of admixture with London and Kinkaid broodstock populations. Sunfish Creek was stocked from 1950 to 1960 with Ohio strain fish, likely originating from New York (Clark 1964). It is currently a self-sustaining population and no longer supplemented. The Little Muskingum River (locality 2) is a natural and self-sustaining population with no history of stocking. There is no evidence that Berlin Lake (locality 1) was ever stocked, and it is also considered a natural and self-sustaining population. In Ohio only 7 of 17 Muskellunge populations are considered self-sustaining and 12 of 17 are introduced (Kerr 2011).

West Virginia is stocked from brood sources that originated primarily from the Chautauqua State Fish Hatchery (NYH, locality 8) (K. Zipfel, West Virginia Division of Natural Resources, personal communication). Prior to 2000, Muskellunge from Kentucky, Ohio, and West Virginia were stocked widely across the state (J. Hansbarger, West Virginia Division of Natural Resources, personal communication). Elk River (locality 5) and Middle Island Creek (locality 6) were stocked extensively prior to 2000 with broodstock from NYH. Both are not currently stocked and are self-sustaining. Populations at the New and Kanawha rivers (NKR; locality 7) were likely derived from NYH stocks and are currently used for stocking throughout the state (K. Zipfel, personal communication). The Buckhannon River (locality 4) is managed with NKR-derived broodstock. In West Virginia, most of the populations are in rivers and 64 out of 110 are considered self-sustaining but only 44 of 110 are indigenous. Given the extensive use of the New York strain of Muskellunge throughout West Virginia waters, we might expect little population structuring.

TABLE 1. Site number, locality information, and sample size (*N*) for the Muskellunge used in this study. Site numbers are shown in Figure 1. Abbreviations are as follows: SFH = State Fish Hatchery.

Site number	Locality	Abbreviation	*N*
1	Berlin Lake, Portage County, Ohio	BER	20
2	Little Muskingum River, Washington County, Ohio	LMR	20
3	Sunfish Creek, Pike County, Ohio	SUN	19
4	Buckhannon River, Upshur County, West Virginia	BR	20
5	Elk River, Kanawha County, West Virginia	ELK	20
6	Middle Island Creek, Pleasants County, West Virginia	MIC	20
7	New and Kanawha rivers, Kanawha County, West Virginia	NKR	15
8	Chautauqua SFH, New York	NYH	16
9	Kincaid SFH, Ohio	KH	20
10	London SFH, Ohio	LH	20
11	Minor Clark SFH, Kentucky	MCH	12

Microsatellite genotyping.— Samples were obtained from Kentucky, New York, Ohio, and West Virginia agency personnel. Genomic DNA was purified from fin clips preserved in ethanol or from dried scales using a DNeasy Tissue kit (Qiagen) from 202 Muskellunge from 11 locations in the Ohio River drainage (Table 1;

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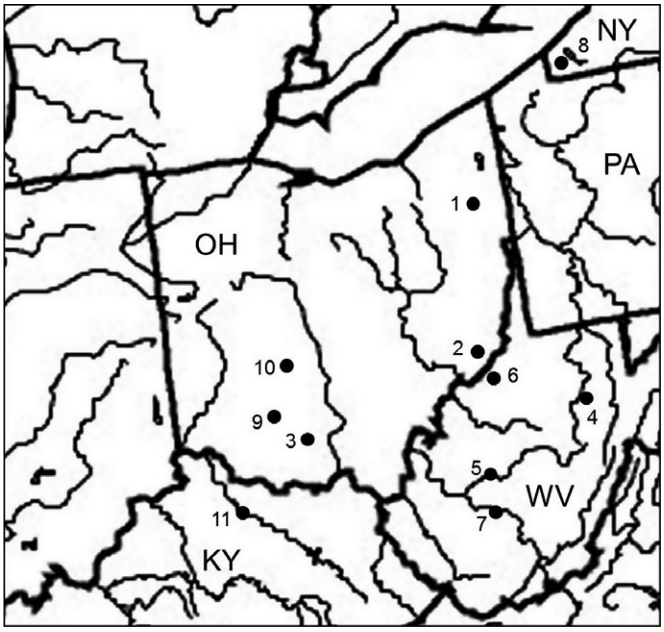

FIGURE 1. Map showing the locations (numbered black dots) of Muskellunge populations sampled for genetic analysis. The numbers refer to the localities identified in Table 1. State abbreviations are as follows: OH = Ohio, NY = New York, PA = Pennsylvania, WV = West Virginia, and KY = Kentucky.

Figure 1). The locations included hatchery or broodstock sources, self-sustaining populations, and supplemented populations. Five tetranucleotide microsatellite loci were surveyed: *B117*, *B304*, and *B451* (Aguilar et al. 2005) and *EmaD5* and *EmaD6* (Sloss et al. 2008). The PCR conditions were the same for all primer pairs: 96°C for 3 min followed by 34 cycles of 94°C for 30 s, 58°C for 30 s, and 72°C for 1 min. Microsatellite fragments were separated using 8% polyacrylamide gel electrophoresis and stained with ethidium bromide. Fragment sizes were determined using a GelDoc photodocumentation system and Quantity One version 4.1.1 software (BioRad). Multiple lanes of a 25-bp ladder were run on each gel to ensure consistent allele size scoring within and between gels.

Allelic diversity and observed and expected heterozygosity were estimated using GenAlEx version 6.3 (Peakall and Smouse 2012). Departures from Hardy–Weinberg equilibrium were tested for each locus in each population using GenAlEx. Pairwise estimates of F_{ST} and their significance, tests of linkage disequilibrium, and an analysis of molecular variance (AMOVA) were performed using Arlequin version 3.5.1.2 (Excoffier and Lischer 2010). The AMOVA was partitioned within and between populations. Significance levels were adjusted for multiple tests by use of the sequential Bonferroni method (Rice 1989).

Genetic clustering.— The program STRUCTURE (Pritchard et al. 2000) was used to determine the number of genetic clusters using 202 individuals and five unlinked microsatellite loci across 11 sampling localities. STRUCTURE operates in a Bayesian framework and maximizes Hardy–Weinberg equilibrium while simultaneously minimizing linkage disequilibrium between user-defined genotypic clusters (K); each model of K is then assigned a likelihood score. Ten independent runs with random starting seeds were used to determine the number of genotypic clusters (K) for values of K ranging from 1 to 11. Each run consisted of a Markov chain–Monte Carlo (MCMC) of 500,000 iterations, with 100,000 iterations discarded as burn-in. The admixture model and correlated allele frequency prior were used to model ancestry, and the admixture parameter, α, was inferred from the data. To detect the uppermost hierarchical population genetic structure, ΔK was calculated using STRUCTURE HARVESTER (Evanno et al. 2005; Earl and von Holdt 2011). Label switching and multimodality were controlled for using CLUMPP (Jakobsson and Rosenberg 2007), and final results were visualized using DISTRUCT (Rosenberg 2003). This procedure was repeated within STRUCTURE clusters to detect substructure.

Gene flow.— To quantify contemporary rates of gene flow (m), BAYESASS (Wilson and Rannala 2003) was used. The program BAYESASS estimates the proportion of migrants per generation over several recent generations. Given a generation time of Ohio River Muskellunge (3–4 years for males, 4–5 years for females; Miles 1978), all gene flow estimates should fall within the last 30 years. The STRUCTURE clusters were used as initial population assignments for BAYESASS. Ten independent simulations with varying starting seeds were run following the recommendations of Faubet et al. (2007) and the best-fit run determined using a Bayesian deviance measure (Spiegelhalter et al. 2002). Each run consisted of a MCMC of 30,000,000 iterations, sampled every 2,250 iterations, with 7,500,000 iterations discarded as burn-in. Chain mixing was optimized in pilot runs to ensure that MCMC acceptance ratios fell within a window of 20% to 40%. This procedure was repeated for each subcluster reported by STRUCTURE. Pilot runs determined that all Δ_A, Δ_F and Δ_M be set to 0.85, 1.00, and 0.70, respectively. The MCMC stationarity for each run was confirmed using TRACER (Rambaut and Drummond 2007).

Historical gene flow ($M = m_h/\mu$) among the genetic clusters recovered by STRUCTURE was estimated using MIGRATE-N (Beerli 2008). The MIGRATE-N estimates gene flow over approximately $4N_e$ generations (thousands of years) for larger populations (Beerli 2009). With potentially smaller Muskellunge populations, the temporal inference would be shallower (e.g., hundreds of years); however, these estimates would still significantly predate contemporary gene flow estimates. Individual runs consisted of a MCMC of 10,000,000 iterations, sampled every

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500 iterations; 5,000,000 iterations were discarded as burn-in. Each analysis was replicated five times, resulting in 50,000 recorded points. Four statically heated chains were used, with temperatures of 1.0, 1.5, 3.0, and 1,000,000. Mutation scaled effective population sizes (θ) for each population were modeled with uniform priors with a lower boundary of 0 and upper boundary of 100. Historical gene flow estimates were modeled with uniform priors, with lower and upper boundaries set to 0 and 3,000. All parameters were deemed to be sufficient if an effective sample size of over 1,000 was observed. This procedure was repeated for each subcluster reported by STRUCTURE.

Temporal change in gene flow (Δm; Chiucchi and Gibbs 2010; Converse et al. 2015) was estimated by subtracting converted historical gene flow estimates from MIGRATE from the contemporary gene flow estimates generated by BAYESASS ($\Delta m = m - m_h$). Multiplying historical gene flow estimates ($M = m_h/\mu$) by a mutation rate, μ, converts estimates into the proportion of migrants per generation (m) used by BAYESASS. Because MIGRATE scales its estimates by mutation rate (Beerli 2008), the mutation rate assumed for conversion is integral to recovering reliable results for comparisons to BAYESASS. A common mutation rate of 5.0×10^{-4} mutations site^{-1} generation^{-1} was used for these conversions (Garza and Williamson 2001; Estoup et al. 2002; Chiucchi and Gibbs 2010). If the mutation rate is higher than 5.0×10^{-4}, the temporal changes in gene flow (see Results) would be underreported. If the mutation rate is lower, the results would be overreported. Positive values of Δm indicate increased contemporary gene flow, negative values indicate reduced contemporary gene flow, and values near zero indicate no temporal changes in gene flow. The Δm was estimated amongst all substructure reported by STRUCTURE.

RESULTS

Microsatellite DNA

Data for five microsatellite loci were obtained for all 202 individuals surveyed. Each locus was polymorphic in each population. The numbers of alleles per locus ranged from 3 to 14 and averaged 8.0 (Table A.1 in the Appendix). The observed heterozygosity within populations averaged across loci ranged from 0.60 to 0.82. The observed heterozygosity and number of alleles for hatchery populations ranged from 0.60 to 0.71 and 6.0 to 8.4, respectively. Values for wild and managed populations were 0.60–0.82 and 7.4–10.0. This is slightly lower but similar to the number of alleles observed in Lake Huron Muskellunge ($\bar{X} = 9.6$ alleles per locus; Wilson et al. 2016). After adjusting for multiple comparisons (adjusted $P = 0.001$), 4 of 55 comparisons identified significant departures from Hardy–Weinberg equilibrium (*EmaD5* and *EmaD6* in locality 4, Buckhannon River, and locality 6, Middle Island Creek). Linkage analysis identified 3 of 50 pairwise comparisons after adjusting for multiple comparisons (adjusted $P = 0.001$). Given that we would expect two significant departures among 50 comparisons, we conclude that the data is consistent with linkage equilibrium. In the AMOVA analysis, 12.4% of the molecular variance was attributed to differences among populations. The remainder (87.6%) was due to variation within populations. Pairwise F_{ST} comparisons ranged from 0.028 to 0.244 (Table A.2). All comparisons were significantly different from zero except the KH–LH comparison ($P > 0.05$).

The program STRUCTURE initially detected two genetic clusters, $\Delta K = 2$ (Figure 2; Supplementary Figures 1, 2 provided in the online version of this article), with low levels of admixture, mostly limited to Sunfish Creek (SUN) and the Elk River (ELK) (Figure 2). Populations in Ohio clustered together as did populations in West Virginia. Substructure was detected in both the Ohio and West Virginia clusters. The Ohio cluster resolved at $\Delta K = 4$ (Figure 2; Supplementary Figures 3, 4) and the West Virginia cluster at $\Delta K = 2$ (Figure 2). Notably, natural populations could be distinguished from stocked or hatchery populations. Within the Ohio cluster, the Kincaid Hatchery (KH) was found to be a discrete genetic cluster, with low levels of admixture found within the population but high levels of admixture in other populations. The London Hatchery (LH) and Minor Clark Hatchery (MCH) showed high levels of admixture, mostly among other hatcheries, with LH receiving a majority of its admixture from KH. As would be expected, all hatchery populations exhibited low levels of admixture coming in from natural populations. Berlin Lake (BER) and the Little Muskingum River (LMR) showed the least amount of hatchery admixture, while Sunfish Creek (SUN) was found to share ancestry with the hatchery populations. This is consistent with Sunfish Creek being one of the source populations for the hatcheries.

Within the West Virginia cluster, all natural populations fell together, with their historic stock population, New York (NYH), falling out as distinct. When the natural populations in West Virginia are analyzed without NYH, STRUCTURE recovers $K = 2$, with Middle Island Creek, a natural population, forming a cluster, and Elk, Buckhannon, and New–Kanawha rivers, the latter two managed populations, composing the second. Middle Island Creek and New–Kanawha rivers show little admixture while the Elk and Buckhannon rivers show moderate levels. Admixture detected at low levels (e.g., within KH; Figure 2) may be a result of resolution due to the number of loci employed (five) (Table 2).

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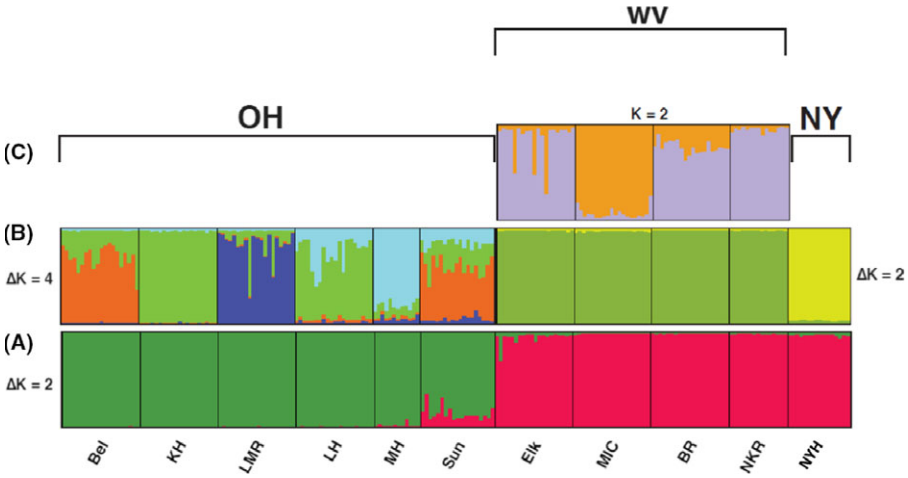

FIGURE 2. Analyses of Muskellunge populations using the program STRUCTURE, showing **(A)** all populations, $\Delta K = 2$; **(B)** Ohio populations, $\Delta K = 4$, and West Virginia and New York populations, $\Delta K = 2$; and **(C)** West Virginia populations, $\Delta K = 2$. Locality abbreviations are as in Table 1. [Color figure can be viewed at afsjournals.org.]

Gene Flow

For historical estimates of gene flow based on coalescence, five loci and sample sizes of 20 are sufficient to detect differences in gene flow (P. Beerli, Florida State University, personal communication). Contemporary estimates may be more sensitive to small sample sizes as they are not coalescent in nature. Based on the results from STRUCTURE, gene flow was estimated at five levels. The first was between Ohio and West Virginia populations, the second between West Virginia populations and New York, and the third was among West Virginia populations (MIC and ELK, BR, NKR). Because of the complex structure within Ohio, gene flow was estimated at two levels. The first assumed four populations (BER–SUN, LMR, MCH, and KH–LH) and the second assumed six (BER, SUN, LMR, MCH, KH, and LH).

Temporal changes in gene flow (Δm) were observed between West Virginia and Ohio, West Virginia and New York, and within West Virginia (Figure 3). Gene flow between Ohio and West Virginia has changed little over time ($-0.01 < \Delta m < -0.005$), as has gene flow between West Virginia and New York ($\Delta m < +0.01$). Gene flow within West Virginia has increased slightly over time ($\Delta m \sim +0.02$). The Δm estimates for gene flow within Ohio for $K = 4$ and $K = 6$ (Figure 4) identified more variation among these routes. At $K = 4$, a majority of gene flow routes exhibited no change to slight increases over time ($\Delta m < +0.03$), with the exceptions of LMR to MCH ($\Delta m = +0.047$), KH–LH to LMR ($\Delta m = +0.061$), and KH–LH to SUN–BER ($\Delta m = +0.080$).

A similar pattern is recovered for $K = 6$ within Ohio (Figure 4). A majority of gene flow routes show no change to slight increases over time ($\Delta m < +0.03$). Several gene flow routes exhibited large increases over time: MCH to LH ($\Delta m = +0.040$), KH to LMR ($\Delta m = +0.042$), LH to KH ($\Delta m = +0.050$), KH to BER ($\Delta m = +0.064$), KH to SUN ($\Delta m = +0.075$), BER to SUN ($\Delta m = +0.081$), and KH to LH ($\Delta m = +0.169$). Most routes with increased levels of contemporary gene flow ($+\Delta m$) are likely associated with stocking in Ohio.

Subtracting historical rates of gene flow from contemporary rates shows that Muskellunge gene flow has changed very little, with an overall mean increase among routes of ~0.01. While the increase is small, several gene flow routes with substantial increases over time seem to be driving it. Gene flow from LH into KH increased by

TABLE 2. Historical (left of the comma) and contemporary (right of the comma) gene flow estimates within Ohio for $K = 4$. Values above the diagonal are for migrants from the row-listed populations to the column-listed populations. Values below the diagonal are for migrants from the column-listed populations to the row-listed populations. See Table 1 for site abbreviations.

Site	SUN–BER	LMR	KH–LH	MCH
SUN–BER		0.005, 0.14	0.005, 0.085	0.004, 0.020
LMR	0.006, 0.023		0.006, 0.067	0.006, 0.017
KH–LH	0.005, 0.014	0.004, 0.010		0.005, 0.035
MCH	0.007, 0.025	0.005, 0.052	0.011, 0.034	

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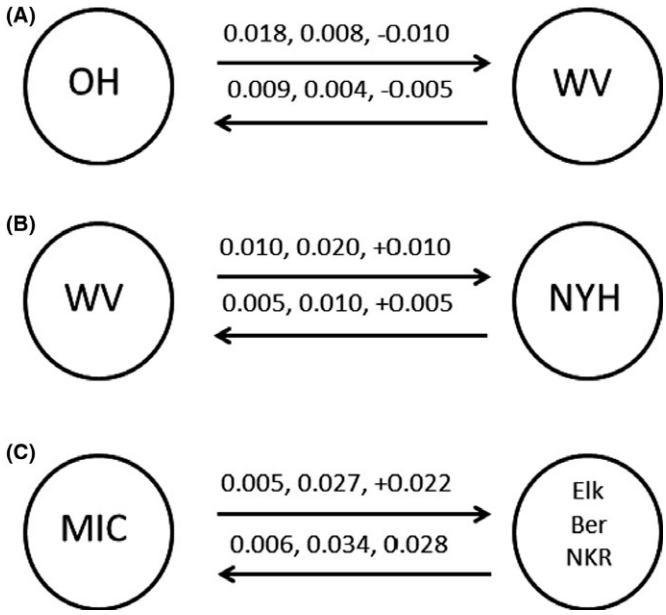

FIGURE 3. Gene flow estimates by population, showing **(A)** gene flow between West Virginia (WV) and Ohio (OH), **(B)** gene flow between West Virginia and New York (NYH), and **(C)** gene flow within West Virginia. The numbers above the arrows indicate gene flow metrics (historical gene flow, contemporary gene flow, Δm). Site abbreviations are defined in Table 1.

0.169, while gene flow from MCH into LH increased by 0.081 (Table 3; Figure 4). The high values may be a reflection of the young age of the hatchery populations. If these two routes, and other large values mentioned above, are removed from the analysis, overall Muskellunge gene flow has remained stable over time, with a slight increase of 0.001. The broodstock source for KH was in part derived from Clearfork Lake, the broodstock source for LH.

DISCUSSION

Despite extensive studies of population structuring in other parts of its range (Miller et al. 2012; Kapuscinski et al. 2013), this is the first study to examine genetic structure among Ohio River Muskellunge populations, a part of the distribution mostly lacking in a recent review of Muskellunge genetics (Miller et al. 2017). We observed levels of variation similar to other studies (Wilson et al. 2016) and significant population differentiation.

Population Structure

A surprising observation in the microsatellite data was that there are two population groups corresponding to west of the Ohio River (localities 1–3, 9–11) and east of the Ohio River (localities 4–8). We recognize two hypotheses that are not mutually exclusive. One, these populations are associated with different management plans and different broodstock sources currently used by the states of Ohio and West Virginia. For example, Scribner et al. (2015) observed that differences among populations of Michigan Muskellunge were a function of similarities among the hatchery strains used. Currently utilized Ohio broodstock sources were derived from streams in southern Ohio (e.g., Sunfish Creek). Thus, Sunfish Creek has made a substantial contribution to stock structure in Ohio

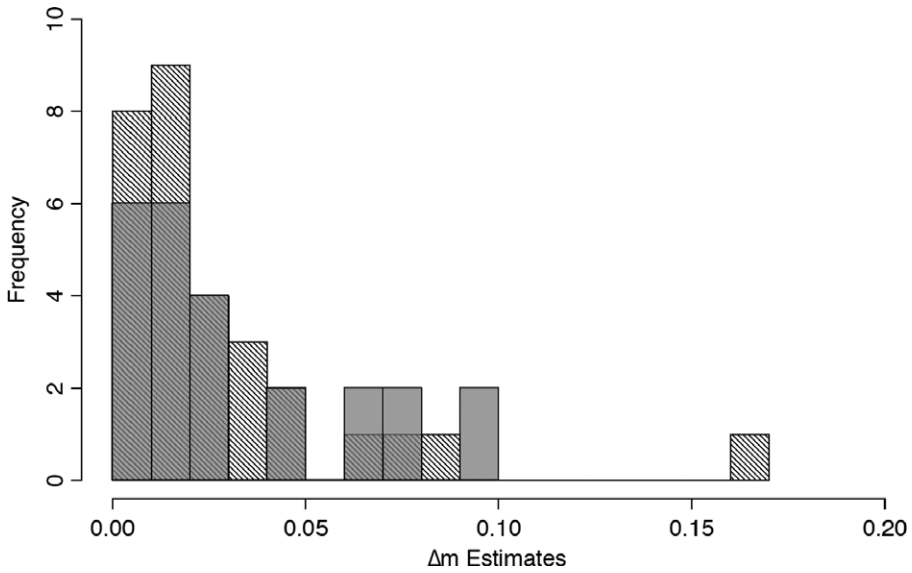

FIGURE 4. Distribution of Δm ($m - m_h$) estimates for gene flow within Ohio. The solid gray histogram denotes Ohio at $K = 4$ and the dashed histogram at $K = 6$.

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TABLE 3. Historical (left of the comma) and contemporary (right of the comma) gene flow estimates within Ohio for $K = 6$. Values above the diagonal are for migrants from the row-listed populations to the column-listed populations. Values below the diagonal are for migrants from the column-listed populations to the row-listed populations.

Site	BER	SUN	LMR	KH	LH	MCH
BER		0.006, 0.019	0.005, 0.014	0.007, 0.071	0.007, 0.029	0.005, 0.015
SUN	0.008, 0.089		0.007, 0.025	0.006, 0.082	0.008, 0.040	0.006, 0.033
LMR	0.007, 0.019	0.0050, 0.016		0.006, 0.048	0.007, 0.029	0.005, 0.015
KH	0.006, 0.015	0.006, 0.014	0.005, 0.013		0.008, 0.058	0.005, 0.016
LH	0.006, 0.020	0.005, 0.016	0.005, 0.016	0.006, 0.178		0.005, 0.045
MCH	0.008, 0.019	0.005, 0.039	0.006, 0.039	0.010, 0.028	0.006, 0.027	

Muskellunge populations and showed admixture with other populations. West Virginia was originally stocked from NYH fish. Several current broodstock sources can claim their origins in New York fish. Despite extensive use of NYH fish, our contemporary gene flow estimates suggested little flow from New York to West Virginia streams. Although gene flow estimates doubled (0.005 to 0.010), they are still low. If the stocked individuals failed to breed (e.g., Buckhannon River) or if our sampling did not fully encompass admixed individuals, then the migration rates recovered would be lower. West Virginia does have a substantial proportion of indigenous (44 out of 110) and self-sustaining (64 out of 110; Kerr 2011) populations that may mask the signature of New York alleles.

The second hypothesis suggests that the Ohio River could be a substantial barrier to gene flow. Muskellunge in this region prefer low-gradient streams with vegetation (Trautman 1981). Locks and dams, flow alterations, and habitat loss along the Ohio River might serve as barriers to gene flow both along and across the river. Muskellunge are uncommon along the main stem of the Ohio River; between 1957 and 2014, only 16 individuals were collected by the Ohio River Sanitation Commission (2014) in their surveys. Only two Muskellunge were reported along the Ohio by Muskellunge fishermen since 2008 (C. Wagner, Ohio Department of Natural Resources, personal communication). A single Muskellunge was reported in a 2012 Ohio River creel survey. Whether this is a reflection of true abundance or the biases associated with sampling protocols used by the Ohio River Sanitation Commission or fishermen is unknown.

Despite the relative close proximity of many of the populations, all pairwise comparisons were significantly different, suggesting limited connectivity. Wilson et al. (2016) observed highly significant differences among populations along the shore of Lake Huron, even sites separated by as little as 20 km. They suggested either site fidelity or philopatry as explanations. If differentiation is possible among populations without obvious barriers to gene flow, structures such as the Ohio River lock and dam system and other impoundments would likely constitute formidable barriers to gene flow. However, Muskellunge stocked in Michigan appear to have spread beyond the stocked localities suggesting that gene flow is possible (Scribner et al. 2015). The historical gene flow estimates also suggest that there was limited connectivity within and between Ohio and West Virginia (Figure 3) and that gene flow has not appreciably increased despite stocking. The MIGRATE estimates of gene flow would reflect conditions well before any anthropogenic habitat alterations consistent with predictions of site fidelity or philopatry (Wilson et al. 2016).

Management

Our microsatellite data identified two groups of populations that may correspond to the management practices of Ohio and West Virginia. Further, we observed significant differentiation even among local populations. Historically, the West Virginia Division of Natural Resources used NYH fish for stocking throughout the state. More recently, fish from the Kanawha and New rivers were used due to a viral hemorrhagic septicemia importation ban (K. Zipfel, personal communication). Middle Island Creek is considered native but was supplemented with fish of Middle Island Creek origin. We recommend continued use of broodstock obtained from West Virginia waters (e.g., MIC, NKR).

Ohio uses two broodstock lakes (Clearfork and Saltfork lakes) and fish from the Licking River (Kentucky). We did observe a decrease in the number of alleles and average heterozygosity from Clearfork to Saltfork stocks. Clearfork Lake fish were used to establish the Saltfork Lake stock. We recommend maintaining this broodstock strategy. The broodstock lake populations should be monitored to ensure no further loss of variation. Consistent with the recommendations of Jennings et al. (2010), we suggest that stocking be avoided in water bodies with natural reproduction (e.g., Sunfish Creek, Little Muskingum River) and that maintenance stocking of populations with no discernable reproduction (e.g., Buckhannon River, West Virginia) use local broodstock sources (e.g., NKR in West Virginia).

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SUPPORTING INFORMATION

Additional supplemental material may be found online in the Supporting Information section at the end of the article.

Appendix: Microsatellite DNA Detailed Data

TABLE A.1. Sample sizes (N), number of alleles (N_a), and observed (H_o) and expected (H_e) heterozygosity evaluated in this study for populations of Muskellunge. Site numbers and populations are from Table 1 and Figure 1.

Site number and population	Locus and average	N	N_a	H_o	H_e
1 Berlin Lake (BER)	*B117*	20	8	0.85	0.80
	EmaD6	20	7	0.75	0.71
	B451	20	10	0.85	0.88
	EmaD5	20	10	0.85	0.79
	B304	20	3	0.20	0.51
	Average		7.4	0.69	
2 Little Muskegon River (LMR)	*B117*	20	5	0.65	0.77
	EmaD6	18	10	0.78	0.83
	B451	18	12	0.94	0.87
	EmaD5	18	13	0.83	0.88
	B304	20	3	0.35	0.63
	Average		8.6	0.71	

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TABLE A.1. Continued.

Site number and population	Locus and average	N	N_a	H_o	H_e
3 Sunfish Creek (SUN)	*B117*	18	12	0.61	0.84
	EmaD6	18	11	0.72	0.84
	B451	15	10	0.53	0.85
	EmaD5	19	14	1.00	0.87
	B304	18	3	0.39	0.59
	Average		10.0	0.65	
4 Buckhannon River (BR)	*B117*	20	11	0.75	0.86
	EmaD6	20	6	0.55	0.68
	B451	20	13	0.85	0.89
	EmaD5	20	9	0.60	0.79
	B304	20	3	0.40	0.33
	Average		8.4	0.63	
5 Elk River (ELK)	*B117*	20	8	0.75	0.82
	EmaD6	19	12	0.68	0.85
	B451	20	11	0.85	0.85
	EmaD5	20	11	0.80	0.84
	B304	20	4	0.85	0.65
	Average		9.2	0.82	
6 Middle Island Creek (MIC)	*B117*	20	9	0.45	0.85
	EmaD6	20	7	0.40	0.63
	B451	20	9	0.85	0.82
	EmaD5	20	9	0.75	0.77
	B304	20	4	0.55	0.62
	Average		7.6	0.60	
7 New and Kanawha rivers (NKR)	*B117*	14	11	0.71	0.89
	EMAD6	15	9	0.47	0.85
	B451	14	9	0.79	0.84
	D5	15	9	0.67	0.81
	B304	15	4	0.47	0.46
	Average		8.4	0.62	
8 Chautauqua Hatchery (NYH)	*B117*	16	10	0.88	0.87
	EMAD6	16	4	0.56	0.68
	B451	16	10	1.00	0.87
	D5	16	11	0.56	0.84
	B304	16	3	0.56	0.61
	Average		7.6	0.71	
9 Kincaid Hatchery (KH)	*B117*	20	6	0.70	0.66
	EmaD6	20	8	0.55	0.77
	B451	20	6	0.80	0.72
	EmaD5	20	8	0.80	0.83
	B304	20	2	0.15	0.49
	Average		6.0	0.60	
10 London Hatchery (LH)	*B117*	20	10	0.90	0.78
	EmaD6	20	10	0.95	0.85
	B451	20	6	0.75	0.60
	EmaD5	20	12	0.50	0.81
	B304	20	4	0.35	0.57
	Average		8.4	0.69	

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TABLE A.1. Continued.

Site number and population	Locus and average	N	N_a	H_o	H_e
11 Minor Clark Hatchery (MCH)	*B117*	12	6	0.50	0.68
	EmaD6	12	6	0.42	0.76
	B451	12	7	0.83	0.75
	EmaD5	12	10	1.00	0.83
	B304	12	3	0.33	0.29
	Average		6.4	0.62	

TABLE A.2. Pairwise F_{ST} values for all comparisons among populations of Muskellunge surveyed in this study. All comparisons are significantly different from zero except KH–LH (marked with an asterisk).

Population	BER	KH	LMR	LH	MCH	SUN	ELK	MIC	BR	NKR	NYH
BER	0.000										
KH	0.074	0.000									
LMR	0.055	0.093	0.000								
LH	0.073	0.028*	0.083	0.000							
MCH	0.126	0.099	0.123	0.084	0.000						
SUN	0.039	0.167	0.065	0.050	0.082	0.000					
ELK	0.119	0.157	0.114	0.146	0.139	0.057	0.000				
MIC	0.146	0.174	0.124	0.149	0.150	0.113	0.078	0.000			
BR	0.224	0.248	0.208	0.230	0.244	0.132	0.100	0.113	0.000		
NKR	0.167	0.179	0.153	0.171	0.174	0.082	0.059	0.094	0.064	0.000	
NYH	0.113	0.138	0.091	0.115	0.111	0.098	0.102	0.107	0.198	0.136	0.000

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